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CICS MENU PROGRAM COMPILE AND LINK JOB SUMMARY

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Student ID: KC03FE1
Assignment: DCMAFD02 Assignment 4
Program Name: UUMNUFE1
Transaction: 9FE1
Date: October 31, 2025

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JOB INFORMATION

Job Name: UUMNUFE1
Job Number: JOB09461
Purpose: Translate, Compile, and Link CICS COBOL menu program
Status: COMPLETED SUCCESSFULLY
Final RC: 0000
System: 8561
Execution Date: October 31, 2025
Execution Time: 23:18:01 - 23:18:05

JOB STATISTICS

Cards Read: 19
SYSOUT Print Records: 1,373
SYSOUT Punch Records: 0
SYSOUT Spool Kbytes: 89
Total Execution Time: 0.06 minutes
Total CPU Time: 0.06 seconds

JOB STEPS SUMMARY

STEP 1: COBOL (Translation & Compilation)

Proc Step: COBOL
Program: IGYCRCTL (Enterprise COBOL Compiler 6.4.0)
Return Code: 0000
CPU Time: 0.05 seconds
Real Time: LESS THAN 0.1 seconds
Region: 0M
Paging: 0 page faults

Compiler Options:

- NODYNAM, RENT
- CICS('COBOL3,SP')
- AMODE(31), RMODE(AUTO)
- OPTIMIZE(0)
- FLAG(I,I)

Source Information:

- Source Records: 416
- Data Division Statements: 123
- Procedure Division Stmts: 57
- Generated COBOL Stmts: 41
- Program Complexity: 98

Messages: No errors or warnings
Status: CLEAN COMPILATION

STEP 2: COPYLINK (Copy DFHEILID)

Proc Step: COPYLINK
Program: IEBGENER
Return Code: 0000
CPU Time: 0.00 seconds

Purpose: Reblock DFHEILID for binder step
Status: SUCCESSFUL

STEP 3: LKED (Link-Edit/Bind)

Proc Step: LKED
Program: IEWL (z/OS Binder)
Return Code: 0000
CPU Time: 0.01 seconds
Region: 0M

Binder Options:

- LIST, XREF, RENT, MAP
- AMODE(31), RMODE(ANY)
- REENTRANT, REUSABLE

Output Library: TSOECCC.CICSTS12.STUDENT.LOADLIB

Member Name: UUMNUFE1
Volume Serial: KCTR05
Disposition: REPLACED
Module Size: 00001CA0 (hex) = 7,328 bytes
DASD Size: 0000E000 (hex) = 57,344 bytes

Module Attributes:

- AMODE: 31
- RMODE: ANY
- RENT: YES (Reentrant)
- REUS: YES (Reusable)
- EXECUTABLE: YES
- ENTRY POINT: UUMNUFE1

PROGRAM DETAILS

Program ID: UUMNUFE1
Author: Lucas Delvoie
Date Written: October 25, 2025
Language: Enterprise COBOL for z/OS 6.4.0

Program Description:

CICS interactive menu program that presents a main menu with multiple choice options. Handles user input and routes to appropriate processing paragraphs based on selection.

WORKING-STORAGE SECTION

Copybook: MNUSFE1
Data Structures:

- MNUMFE1I (Input map structure)
- MNUMFE1O (Output map structure)

LINKAGE SECTION

DFHCOMMAREA: PIC X (1 byte)

PROCEDURE DIVISION LOGIC

000-MAIN:

- Sets up MAPFAIL condition handler
- Receives map input from terminal
- Routes to 200-MAIN-LOGIC

100-FIRST-TIME:

- Handles initial screen display (MAPFAIL condition)
- Initializes output area with LOW-VALUES
- Sends map with ERASE option
- Returns control with TRANSID('9FE1')

200-MAIN-LOGIC:

- Evaluates CHOICEI field
- Routes to appropriate processing paragraph:
 - * Choice '1' → 300-CHOICE-ONE
 - * Choice '2' → 400-CHOICE-TWO
 - * Choice '3' → 500-CHOICE-THREE
 - * Choice '4' → 600-CHOICE-FOUR
 - * Choice '9' → 900-EXIT
 - * Other → 800-INVALID-INPUT

300-CHOICE-ONE:

- Displays: "Choice 1 is not available"
- Returns with TRANSID('9FE1')

400-CHOICE-TWO:

- Displays: "Choice 2 is not ready "
- Returns with TRANSID('9FE1')

500-CHOICE-THREE:

- Displays: "Choice 3 is being tested "
- Returns with TRANSID('9FE1')

600-CHOICE-FOUR:

- Displays: "Choice 4 is not working "
- Returns with TRANSID('9FE1')

800-INVALID-INPUT:

- Displays: "Invalid input - KC03FE1"
- Returns with TRANSID('9FE1')

900-EXIT:

- Displays: "APPLICATION ENDING"
- Returns control to CICS (no TRANSID)

CICS COMMANDS USED

1. EXEC CICS HANDLE CONDITION

- MAPFAIL(100-FIRST-TIME)

2. EXEC CICS RECEIVE MAP

- MAP('MNUMFE1')
- MAPSET('MNUSFE1')
- INTO(MNUMFE1I)

3. EXEC CICS SEND MAP

- MAP('MNUMFE1')
- MAPSET('MNUSFE1')
- FROM(MNUMFE1O)
- ERASE (on first-time display)

4. EXEC CICS RETURN

- TRANSID('9FE1') (pseudo-conversational)
- (no TRANSID for final exit)

DATA CROSS-REFERENCE

Key Fields Used:

- CHOICEI: User input field (1 character) - READ
- MSGO: Message output field (40 chars) - MODIFIED
- MNUMFE1O: Output map structure - MODIFIED
- MNUMFE1I: Input map structure - READ

LIBRARY ALLOCATIONS

COBOL Step:

- STEPLIB: IGY640.SIGYCOMP
DFH610.CICS.SDFHLOAD
- SYSLIB: DFH610.CICS.SDFHCOB
KC03FE1.DCMAFD02.A4.MNU.COBOL
KC03FE1.DCMAFD02.COPYBOOK.WS.COBOL
KC03FE1.DCMAFD02.COPYBOOK.PR.COBOL
DFH610.CICS.SDFHMAC
DFH610.CICS.SDFHSAMP
- SYSIN: KC03FE1.DCMAFD02.A4.MNU.COBOL(UUMNUFE1)

LKED Step:

- SYSLIB: DFH610.CICS.SDFHLOAD
CEE.SCEELKED
- SYSMOD: TSOECCC.CICSTS12.STUDENT.LOADLIB

CICS INSTALLATION REQUIREMENTS

1. Program Definition:

- Program: UUMNUFE1
- Language: COBOL
- Status: ENABLED
- Location: TSOECCC.CICSTS12.STUDENT.LOADLIB

2. Transaction Definition:

- Transaction: 9FE1
- Program: UUMNUFE1
- Status: ENABLED
- Description: Assignment 4 Menu Program

3. Mapset Definition:

- Mapset: MNUSFE1
- Status: INSTALLED

TESTING INSTRUCTIONS

1. Ensure CICS region is active
2. Clear screen and enter transaction: 9FE1
3. Press ENTER to display menu
4. Test each menu option:
 - Enter 1 and press ENTER (Choice 1)
 - Enter 2 and press ENTER (Choice 2)
 - Enter 3 and press ENTER (Choice 3)
 - Enter 4 and press ENTER (Choice 4)
 - Enter 9 and press ENTER (Exit)
 - Enter invalid character (test error handling)
5. Verify appropriate messages display for each choice
6. Verify transaction ends properly on choice 9

EXPECTED BEHAVIOR

First Time:

- Menu displays with all choices visible
- Cursor positioned at CHOICE input field
- All fields properly formatted

Valid Choice (1-4):

- Appropriate message displays in MSG field
- Screen refreshes with menu still visible
- Transaction remains active (pseudo-conversational)

Exit (9):

- "APPLICATION ENDING" message displays
- Transaction terminates
- User returned to blank screen

Invalid Input:

- "Invalid input - KC03FE1" message displays
- Menu remains visible
- User can try again

PROGRAM CHARACTERISTICS

- Pseudo-conversational design (efficient CICS resource usage)
- Reentrant and reusable
- Menu-driven interface
- Input validation
- Error handling via HANDLE CONDITION
- Standard CICS command-level programming
- Uses symbolic maps for terminal I/O

FILES CREATED/MODIFIED

Source File: KC03FE1.DCMAFD02.A4.MNU.COBOL(UUMNUFE1)
 Load Module: TSOECCC.CICSTS12.STUDENT.LOADLIB(UUMNUFE1)
 Copybook: MNUSFE1 (generated from BMS)
 Job Output: KC03FE1.UUMNUFE1.JOB09461.D0000101-104

NOTES

- Program successfully compiled with no errors or warnings
- All CICS translator options properly set
- Module is reentrant and suitable for CICS execution
- Program follows pseudo-conversational design pattern
- Uses RETURN with TRANSID for screen redisplay
- Properly handles MAPFAIL condition for first-time execution
- All SEND MAP commands include FROM clause
- RECEIVE MAP command includes INTO clause
- Low-values initialization ensures clean screen display

ISSUES RESOLVED

1. Initial Issue: Mapset/Map naming confusion
 Resolution: Corrected MAP name to 'MNUMFE1' (not 'MNUSFE1')
2. Initial Issue: Missing INTO/FROM clauses
 Resolution: Added INTO(MNUMFE1I) to RECEIVE MAP
 Added FROM(MNUMFE10) to all SEND MAP commands
3. Initial Issue: BMS file had YYY placeholders
 Resolution: Replaced YYY with FE1 throughout BMS source

COMPILATION SUMMARY

- ✓ Translation: Successful (CICS commands expanded)
- ✓ Compilation: Successful (No errors, no warnings)
- ✓ Link-Edit: Successful (Module created)
- ✓ Load Module: Ready for CICS execution
- ✓ Return Code: 0000 (All steps successful)

STATUS: READY FOR CICS INSTALLATION AND TESTING

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END OF MENU PROGRAM JOB SUMMARY

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